

Direct numerical simulation of three-dimensional particle-laden thermal convection using the Lattice Boltzmann Method

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ABSTRACT

In thermal multiphase flows, the modulation in the heat transfer rate due to the presence of finite-size solid particles attract growing interest in recent years. This study focuses on developing a robust and efficient simulation method for particle-laden Rayleigh-Bénard convection. The present work utilizes the double distribution function-based thermal lattice-Boltzmann method (TLBM), which enables successful simulations of multiphase fluid-thermal interactions. The no-slip boundary of the moving solid particles is handled by the interpolated bounce-back scheme. Additionally, Galilean invariant momentum exchange and heat exchange approaches are employed for hydrodynamic force and heat transfer calculation at the solid boundaries. The accuracy of the current method is verified with several benchmark cases, including three-dimensional single-phase Rayleigh-Bénard convection, as well as the settling of hot and cold spherical particles in a three-dimensional enclosure. Furthermore, we explore the modulation of Rayleigh-Bénard flows due to the presence of freely moving finite-size particles. The simulations are conducted on a distributed memory cluster with a 3D domain decomposition technique facilitated by the MPI library. A brief discussion on the parallel performance of the simulations is provided for both particle-laden and single-phase flow scenarios. Finally, the effects of finite-size solid particles on the overall heat transfer efficiency and modulation to the flow field in three-dimensional particle-laden turbulent Rayleigh-Bénard convection are discussed. The results show that addition of solid particles results in a moderate increase in the overall Nusselt number, and this enhancement is mainly due to the increased heat flux transported by the particles.

1. Introduction

In the field of fluid dynamics, thermal particle-laden flows represent a fascinating and complex area of study, intertwining the principles of thermodynamics, fluid mechanics, and particle dynamics. Thermal particle-laden flows are of great importance in numerous industrial and natural processes, ranging from the formation of atmospheric aerosols [1,2] to the design of high-performance combustion systems [3]. The primary challenge in understanding thermal particle-laden flows lies in the intricate interactions between the particles and the fluid medium, further complicated by the influence of thermal gradients.

Nowadays, computational fluid dynamics (CFD) has emerged as a powerful tool in this field, allowing for detailed simulations that capture the subtle and complex interactions between particles and fluid flow. At the same time, due to the progress in numerical methods and computer hardware, these models must be continually refined and validated against experimental data to ensure accuracy and reliability. For example, the development of efficient numerical methods

capable of resolving fluid flows around moving or deforming (solid or flexible) objects led to wider applicability of so-called Interface Resolved (IR) Direct Numerical Simulations (DNS) where the local flow features at the inter-phase interface scales are resolved explicitly. The most common numerical methods used for such IR simulations are the Immersed Boundary Method (IBM) [4,5], Volume-of-Fluid (VoF) [6] and cut-cell [7] based methods.

Subsequently, many works were introduced with extensions and improvements to these methods to handle thermal coupling in addition to the momentum coupling. Yu et al. [8] combined Discrete Lagrange Multiplier-Finite Difference Method to study two-dimensional particle-laden flows with heat transfer. Storm [9] proposed a multiphase DNS method capable of resolving the motion of solid particles coupled with heat transfer effects using the VoF method. Wang et al. [10] introduced an extended version of the so-called Physalis method [11] for thermal flows around spherical particles. In this approach, flow field and thermal effects are theoretically modeled near the particle surface, while

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full Navier–Stokes and energy equations are solved in other regions. It was shown that the thermal Physalis method could be accurately applied to spherical particles with lumped capacitance approximation for Rayleigh number up to 2×10^6 . An important aspect of heat transfer modes in the gas-solid medium was studied by Sun et al. [12], where IR-DNS was used to quantify heat transfer regimes through Nusselt number correlations with the slip-velocity particle Reynolds number and solid-phase volume fraction. Although this work used configuration with fixed, constant temperature particles in uniform flow, the results obtained indicated improved Nusselt number correlations over existing models based on experimental fitting. Multi-physics heat transfer modes for reactive fluid-particle systems were studied in [13] using the ghost-cell based IBM. Flow modulation in the presence of moving, fixed temperature solid particles were studied by Xia et al. [14] where IR-DNS was performed using ghost cell IBM.

Complementary to the traditional CFD solvers, methods based on the kinetic approach such as the Lattice Boltzmann Method (LBM) emerged as powerful alternative simulation tools, thanks to the active development in that area and progress in computational methods and computer systems. Some distinct features of LBM, such as explicit nature of the time advancement scheme, allows to efficiently use modern multi-core and multi-thread computer systems. Another appealing feature is the possibility to locally treat curved solid boundaries with a second-order accuracy, very helpful in treating complex flows laden with solid objects.

Initial developments in the application of LBM to thermal problems with moving spherical solids were presented by Hashemi et al. [15] and by Eshghinejadfard and Thévenin [16]. Hashemi and colleagues presented three-dimensional thermal LBM approach to study the fluid-particle interaction and its modulation on heat transfer during gravitational settling in an enclosure. Adopted scheme with linear interpolation for fluid boundary nodes and extrapolation procedure for temperature boundary conditions were shown to provide good results. Eshghinejadfard and Thévenin [16] combined LBM with IBM and demonstrated applications for catalyst particulate flows. Suzuki et al. [17] extensively validated their implementation of two-dimensional LBM-IBM method. While only a first-order accuracy was observed in benchmark solution, overall agreement with known data from the literature was good. Authors further applied their thermal LBM to study the simplified ice-slurry flow in a channel.

Although the IBM is widely used to treat the fluid-solid interface, the interpolation function used in IBM cannot guarantee the second-order accuracy for general flow and geometries [18]. It is also found that IBM would under-estimate the local dissipation near the particle surface [19]. The bounce-back scheme provides an alternative way to treat the solid boundaries from the mesoscopic perspective, and the force and torque exerted on the particle can be obtained by summing up all the momentum exchanges over the particle surface and then using the impulse momentum theorem. The original simple bounce-back scheme introduces “staircase” effect [20] for curved boundaries, reducing the order of accuracy, but the further improved version called interpolated bounce-back [21,22] offers the second-order accuracy locally and globally. These features make the LBM-IBB method an powerful tool in treating complex boundaries and simulating finite-size particle-laden flows.

The basic idea of interpolated bounce-back for momentum exchange can also be used to derive the thermal boundary conditions for thermal LBM. Considering the special case where the distance from the boundary wall to the nearest lattice node is equal to half of the lattice spacing, Yoshida and Nagaoka [23] proposed a D3Q7 (in LBM notation: three spatial dimensions with seven discrete velocity directions) discrete velocity model (DVM) thermal LBM model with an Multiple Relaxation Time (MRT) collision operator, for the Dirichlet and Neumann boundary conditions. An asymptotic analysis was also given in their work, showing that second-order accuracy was obtained in both cases. Mezrhab et al. [24] proposed a numerical scheme to solve

the flow field and temperature field using the D2Q9 MRT model and the D2Q5 MRT model respectively, and the bounce-back scheme was implemented to treat Dirichlet boundary conditions. Kruggel-Emden et al. [25] performed extensive validations of the particle packing simulations using thermal IBB coupled with mixed MRT and Single-Relaxation Time (SRT, also known as the BGK collision kernel) collision models. Their work demonstrated applicability of combined thermal LBM-IBB approach to many problems previously studied using traditional Navier–Stokes equation based solvers. Rosemann et al. [26] compared various techniques necessary for the thermal lattice Boltzmann method to handle moving particles in fluid flows when heat transfer was included. These techniques include a new efficient heat transfer evaluation for curved boundaries, a new interpolated bounce-back scheme for handling thermal boundary conditions, and a refill algorithm for temperature distribution functions.

While the above mentioned examples and applications indicate good efficiency of application of LBM-IBB to thermal flows, they were focused on laminar cases, and the applicability and stability of thermal LBM-IBM for high Rayleigh number flows have not been validated yet.

Related to performance considerations, recently, Xu and Li [27] demonstrated computationally efficient implementation of thermal LBM for interface-resolved two-dimensional simulation of particulate flows on multi-GPU systems, where relatively high efficiency of 77.5% was achieved. Such high parallel efficiency results indicate that explicit nature of discrete form of LBM evolution can be used efficiently on modern parallel systems.

This paper aims to study thermal particle-laden flows using the lattice Boltzmann method with fully resolved solid-fluid interfacial hydrodynamics and thermal interactions. To achieve this, first we present implementation details of the three-dimensional LBM-IBB approach for fluid and temperature fields. Following extensive validations of the implementation, the method is applied to study particle-laden thermal convection at a relatively high Rayleigh number ($Ra = 10^7$). Obtained results demonstrate robustness of the LBM-IBB approach for thermal flows.

The paper is structured as follows. First, in Sections 2 and 3, governing equations with the details of the numerical method are presented. Then in Section 4, several benchmark solutions were presented to validate our implementation. In Section 5, the case of particle-laden thermal convection with vertical temperature gradient is simulated and discussed. Main conclusions are summarized in Section 6.

2. Problem formulation

The physical problem of the coupled momentum and heat transfer between fluid and submerged solid phase is governed by the Navier–Stokes equations for the fluid phase (Eqs. (1)–(1c)) and Newton and heat transfer equations for the solid phase (Eqs. (2)–(4)). The fluid flow is considered to be incompressible, and the incompressible flow equations based on the Oberbeck-Boussinesq approximation [28,29] are used:

$$\nabla \cdot \mathbf{u}_f = 0, \quad (1a)$$

$$\frac{\partial \mathbf{u}_f}{\partial t} + \mathbf{u}_f \cdot \nabla \mathbf{u}_f = -\frac{1}{\rho_f} \nabla p_m + \nu_f \nabla^2 \mathbf{u}_f - \beta_f (T_f - T_0) \mathbf{g}, \quad (1b)$$

$$\frac{\partial T_f}{\partial t} + \mathbf{u}_f \cdot \nabla T_f = \alpha_f \nabla^2 T_f, \quad (1c)$$

where the subscript f stands for fluid phase, $\mathbf{u}_f$ is the fluid velocity field, $p_m \equiv p + \rho_f g z$ is the modified pressure with p being the pressure, T_f is the fluid temperature, T_0 is the reference temperature, and $\mathbf{g}$ is the gravity vector, ρ_f , ν_f , β_f , and α_f represent the fluid density, kinematic viscosity, thermal expansion coefficient, and thermal diffusivity, respectively.

For the solid phase, freely-moving spherical particles with large in-solid thermal diffusion are considered. The diameter and density of all the particles are the same, and they maintain the same density as the fluid. The governing equations for these particles are the Newton-Euler equations for rigid body translation and rotation, and the heat transfer equation for the solid-phase temperature:

$$m_p \frac{d\mathbf{u}_p}{dt} = \mathbf{F}^H + \mathbf{F}^C + \mathbf{F}^G, \quad (2)$$

$$\frac{d(\mathbf{I} \cdot \boldsymbol{\omega}_p)}{dt} = \boldsymbol{\Gamma}^H, \quad (3)$$

$$m_p c_{p,p} \frac{dT_p}{dt} = \rho_f c_{p,f} \Phi^H, \quad (4)$$

in which m_p and $\mathbf{I}$ represent the mass and moment of inertia of the particle, $c_{p,p}$ and $c_{p,f}$ are the heat capacity of the particle and the fluid, and $\mathbf{u}_p$ and $\boldsymbol{\omega}_p$ represent the translational velocity and the angular velocity of the particle, respectively. The effective body force $\mathbf{F}^G = m_p(1 - \rho_f/\rho_p)\mathbf{g}$ represents the difference between the gravity body force and the buoyancy force, in which ρ_p represents the particle density, all other hydrodynamic forces exerted by the fluid on particle is denoted as $\mathbf{F}^H$, and the contact forces $\mathbf{F}^C$ represents force resulting from the inter particle and particle-wall collisions. $\boldsymbol{\Gamma}^H$ represents the torque exerted on the particles by the fluid and T_p is the particle temperature, while Φ^H represents the heat transfer from the surrounding fluid to the particle. In Eq. (4) the particle temperature is assumed to be uniform within a given particle, which is a simplified treatment when the thermal diffusivity within the solid phase is assumed to be large [10].

3. Numerical method

3.1. Double distribution lattice Boltzmann model

This study uses the single relaxation time lattice Boltzmann method to simulate the velocity and temperature field [30,31], where two distribution functions are introduced to track the evolution of the fluid velocity and temperature:

$$f_\alpha(\mathbf{x} + \mathbf{e}_\alpha \Delta t, t + \Delta t) - f_\alpha(\mathbf{x}, t) = -\frac{1}{\tau_f} [f_\alpha(\mathbf{x}, t) - f_\alpha^{eq}(\mathbf{x}, t)] + \Delta t F_\alpha, \quad (5)$$

$$g_\alpha(\mathbf{x} + \mathbf{e}_\alpha \Delta t, t + \Delta t) - g_\alpha(\mathbf{x}, t) = -\frac{1}{\tau_g} [g_\alpha(\mathbf{x}, t) - g_\alpha^{eq}(\mathbf{x}, t)],$$

where $\mathbf{x}$ denotes the spatial location, $\mathbf{e}_\alpha$ is particle velocity, Δt is the lattice time step, τ_f is the momentum relaxation time, τ_g is the internal energy relaxation time and F_α is the external force component related to the buoyancy force in the α th discrete velocity direction, $\alpha = 0, 1, \dots, 26$. The equilibrium distribution functions for this model are:

$$f_\alpha^{eq} = \rho_f \omega_\alpha \left[1 + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right], \quad (6)$$

$$g_\alpha^{eq} = T \omega_\alpha \left[1 + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right].$$

where c_s represents the lattice sound speed and is equal to $1/\sqrt{3}$ in lattice unit.

The second-order accurate approach to handle the external force term, proposed by Guo et al. [32], is adopted:

$$F_\alpha(\mathbf{x}, t) = \omega_\alpha \left(1 - \frac{1}{2\tau_f} \right) \left[\frac{\mathbf{e}_\alpha \cdot \mathbf{F}}{c_s^2} + \frac{(\mathbf{F} \cdot \mathbf{u} + \mathbf{u} \cdot \mathbf{F}) : (\mathbf{e}_\alpha \mathbf{e}_\alpha - c_s^2 \mathbf{I})}{2c_s^4} \right]. \quad (7)$$

In Eq. (7), the external force $\mathbf{F}$ is given by $-\rho_f g \beta (T_f - T_0)$, where ρ_f is the fluid density and $\mathbf{g}$ is the acceleration due to gravity.

The D3Q27 model was used and $\mathbf{e}_\alpha$ represents the discrete particle velocities, given as:

$$\mathbf{e}_\alpha = \begin{cases} (0, 0, 0), & \alpha = 0, \\ (\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1), & \alpha = 1 - 6, \\ (\pm 1, \pm 1, 0), (0, \pm 1, \pm 1), (\pm 1, 0, \pm 1), & \alpha = 7 - 18, \\ (\pm 1, \pm 1, \pm 1), (\pm 1, \pm 1, \pm 1), (\pm 1, \pm 1, \pm 1), & \alpha = 19 - 26, \end{cases} \quad (8)$$

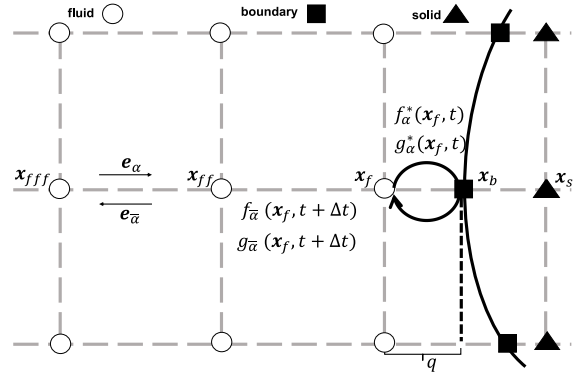


Fig. 1. Boundary treatment in interpolated bounce-back method. Open circles represent fluid nodes, filled triangles represent nodes internal to solid particle and filled squares represent intersection points of curved boundary with discrete velocities $\mathbf{e}_\alpha$ and $\mathbf{e}_{\bar{\alpha}}$. The node $\mathbf{x}_f$ denotes near boundary node, and $\mathbf{x}_b$ denotes the wall position.

with the weighting factors ω_α for D3Q27 model being determined as follows:

$$\omega_\alpha = \begin{cases} \frac{8}{27}, & \alpha = 0, \\ \frac{2}{27}, & \alpha = 1 - 6, \\ \frac{1}{54}, & \alpha = 7 - 18, \\ \frac{1}{216}, & \alpha = 19 - 26. \end{cases} \quad (9)$$

The macroscopic hydrodynamic variables, including density, momentum, and temperature, are obtained through moments of the discrete velocities:

$$\rho_f = \sum_\alpha f_\alpha, \quad p_m = \rho_f c_s^2 = \frac{\rho_f}{3},$$

$$\rho_f \mathbf{u}_f = \sum_\alpha f_\alpha \mathbf{e}_\alpha + \frac{\Delta t}{2} \mathbf{F}, \quad (10)$$

$$T_f = \sum_\alpha g_\alpha.$$

Through Chapman-Enskog expansion, the macroscopic equations (Eq. (1)) can be recovered from the moments of the lattice Boltzmann equations (Eq. (5)). (See Appendix A for the details.) The kinematic viscosity ν , and the thermal diffusivity α of the fluid are related to the relaxation time as:

$$\nu_f = c_s^2 \left(\tau_f - \frac{1}{2} \right) \Delta t, \quad (11)$$

$$\alpha_f = c_s^2 \left(\tau_g - \frac{1}{2} \right) \Delta t.$$

3.2. Implementation of boundary conditions

After performing the collision operator in Eq. (5), fluid particles are streamed, but the distributions for incoming fluid particles $\mathbf{x}_f$ propagated from the solid node $\mathbf{x}_s$ are not available. In the case of standard bounce-back, the solid wall is assumed to be located exactly half way between the nodes $\mathbf{x}_f$ and $\mathbf{x}_s$ (Fig. 1). That way, particles leaving the node $\mathbf{x}_f$ in the direction of $\mathbf{e}_\alpha$ are reflected or “bounced-back” and return to the node $\mathbf{x}_f$, and they move in the direction of the velocity vector opposite to $\mathbf{e}_\alpha$, denoted by $\mathbf{e}_{\bar{\alpha}}$, after the “bounced-back”, leading to the relation:

$$f_{\bar{\alpha}}(\mathbf{x}_f, t + \Delta t) = f_\alpha^*(\mathbf{x}_f, t), \quad (12)$$

where $f_\alpha^*(\mathbf{x}_f, t)$ represents the distribution after collision. Although the “bounce back” scheme is widely accepted, the rigorous derivation of the “bounce-back” scheme is usually not presented in the literature. The distribution function can be divided into the equilibrium part and the non-equilibrium part, and the non-equilibrium part can be explicitly

expressed using the Chapman-Enskog expansion. This explicit expression of the distribution is called the lifting relation, which relates the distribution function to the macroscopic variables. As the macroscopic variables or the gradients of these quantities at the wall are known, the unknown distributions can be constructed using the lifting relation (See Appendix B for details). In more general situations, the moving curved boundaries do not coincide with computational fluid nodes as indicated in Fig. 1, and simple no-slip boundary treatment, such as standard bounce-back, will not give the second-order accurate results. To overcome this, Bouzidi et al. [21] proposed the interpolated bounce-back scheme: (1) when $q \leq 0.5$, a pre-streaming node $\mathbf{x}^{(I)}$ (located between $\mathbf{x}_f$ and $\mathbf{x}_{ff}$) is identified, such that a particle initiated from $\mathbf{x}^{(I)}$ will reach $\mathbf{x}_f$ after bounce-back, $f_a^*(\mathbf{x}^{(I)}, t)$ is interpolated first, then the bounce-back is performed to obtain $f_{\bar{a}}(\mathbf{x}_f, t + \Delta t)$; (2) when $q > 0.5$, bounce-back of $f_a^*(\mathbf{x}_f, t)$ is performed first to obtain $f_{\bar{a}}(\mathbf{x}^{(II)}, t + \Delta t)$, where $\mathbf{x}^{(II)}$ is now located between $\mathbf{x}_f$ and $\mathbf{x}_b$, then interpolation between $\mathbf{x}^{(II)}$ and $\mathbf{x}_{ff}$ ($\mathbf{x}_{fff}$) is used to obtain $f_{\bar{a}}(\mathbf{x}_f, t + \Delta t)$. The interpolation can use two or three points. When two points are used, the linear interpolated bounce-back becomes:

for $q \leq 0.5$:

$$f_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = 2q f_a^*(\mathbf{x}_f, t) + (1 - 2q) f_a^*(\mathbf{x}_{ff}, t) + \Delta f_b, \quad (13)$$

for $q > 0.5$:

$$f_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = \frac{1}{2q} (f_a^*(\mathbf{x}_f, t) + \Delta f_b) - \frac{(1 - 2q)}{2q} f_a^*(\mathbf{x}_f, t),$$

When three points are used, the quadratic interpolated bounce-back can be derived:

for $q \leq 0.5$:

$$f_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = q(2q + 1) f_a^*(\mathbf{x}_f, t) + (1 - 4q^2) \times f_a^*(\mathbf{x}_{ff}, t) - q(1 - 2q) f_a^*(\mathbf{x}_{fff}, t) + \Delta f_b,$$

for $q > 0.5$:

$$f_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = \frac{1}{2q^2 + q} (f_a^*(\mathbf{x}_f, t) + \Delta f_b) + \frac{2q - 1}{q} \times f_a^*(\mathbf{x}_f, t) - \frac{(2q - 1)}{1 + 2q} f_a^*(\mathbf{x}_{ff}, t), \quad (14)$$

where the boundary term is $\Delta f_b = 2\omega_a \rho_f \mathbf{u}_b \cdot \mathbf{e}_{\bar{a}} / c_s^2$. In the above equations, $\bar{a}$ denotes the bounce-back direction, $f_a^*(\mathbf{x}_f, t)$ represents the post-collision distribution, $\mathbf{u}_b$ is the velocity of the curved boundary, and $\mathbf{u}_b = \mathbf{u}_p + \boldsymbol{\omega}_p \times (\mathbf{x}_b - \mathbf{x}_p)$ with $\mathbf{x}_p$ the center of the spherical particle.

In the actual implementation, quadratic, linear or simple bounce-back method is selected depending on the actual number of available fluid nodes.

For the temperature field, the isothermal boundary is achieved using the anti-bounce-back scheme, and the corresponding derivation is provided in Appendix B. The linearly interpolated version of the anti-bounce-back scheme is: [23,33]:

for $q \leq 0.5$:

$$g_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = -(2q g_a^*(\mathbf{x}_f, t) + (1 - 2q) g_a^*(\mathbf{x}_{ff}, t)) + \Delta g_b, \quad (15)$$

for $q > 0.5$:

$$g_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = \frac{1}{2q} (-g_a^*(\mathbf{x}_f, t) + \Delta g_b) - \frac{(1 - 2q)}{2q} g_a^*(\mathbf{x}_f, t),$$

with the boundary term being:

$$\Delta g_b = 2\omega_a T_b, \quad (16)$$

where T_b is the temperature of the wall.

The single node anti-bounce back scheme is used when no additional fluid nodes are available for interpolation purposes:

$$g_{\bar{a}}(\mathbf{x}_f, t + \Delta t) = -g_a^*(\mathbf{x}_f, t) + 2\omega_a T_b. \quad (17)$$

The adiabatic boundary is realized using the bounce back of the distribution g_a in the mesoscopic perspective, which is similar to the bounce back of f_a , and this treatment follows from the lifting relation of g .

3.3. Force, torque and heat exchange of the particles

To account for the force and torque exerted by the fluid on a solid particle, Galilean invariant momentum exchange method introduced by Wen et al. [34] is used:

$$\begin{aligned} \mathbf{F}^H \Delta t &= \sum_{\mathbf{x}_{f,a}} [f_a^*(\mathbf{x}_f, t)(\mathbf{e}_{\bar{a}} - \mathbf{u}_b) - f_{\bar{a}}(\mathbf{x}_f, t + \Delta t)(\mathbf{e}_i - \mathbf{u}_b)], \\ \mathbf{I}^H \Delta t &= \sum_{\mathbf{x}_{f,a}} (\mathbf{x}_b - \mathbf{x}_p) \\ &\quad \times [f_a^*(\mathbf{x}_f, t)(\mathbf{e}_{\bar{a}} - \mathbf{u}_b) - f_{\bar{a}}(\mathbf{x}_f, t + \Delta t)(\mathbf{e}_a - \mathbf{u}_b)]. \end{aligned} \quad (18)$$

After obtaining the hydrodynamic force and torque acting on the particle, the particle velocity and angular velocity can be updated, as well as the particle position and angle.

Similarly to hydrodynamic effect, the heat transfer from fluid to solid particles can be obtained by the heat exchange method [26]:

$$\Phi^H \Delta t = \Delta x^3 \sum_{\mathbf{x}_{f,a}} [g_a^*(\mathbf{x}_f, t) - g_{\bar{a}}(\mathbf{x}_f, t + \Delta t)], \quad (19)$$

and then the particle temperature can be updated accordingly.

3.4. Particle-particle and particle-wall interaction

In the simulation of finite-size particles, particles may approach each other and collide. Therefore, the near-field lubrication interaction and the contact-collision between two particles need to be modeled. When the distance between particles is very close and there are no fluid lattice points, the lubrication force acting on the i th-particle due to its interaction with the j th-particle is modeled using the repulsive force model [35,36]:

$$F_{ij}^P = \begin{cases} 0, & \text{for } |\mathbf{x}_i - \mathbf{x}_j| > R_i + R_j + \zeta; \\ \frac{c_F}{\epsilon_p} \left(\frac{|\mathbf{x}_i - \mathbf{x}_j| - R_i - R_j - \zeta}{\zeta} \right)^2 \left(\frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|} \right), & \text{for } R_i + R_j < |\mathbf{x}_i - \mathbf{x}_j| \leq R_i + R_j + \zeta; \\ \left[\frac{c_F}{\epsilon_p} \left(\frac{|\mathbf{x}_i - \mathbf{x}_j| - R_i - R_j - \zeta}{\zeta} \right)^2 + \frac{c_F}{E_p} \frac{R_i + R_j - |\mathbf{x}_i - \mathbf{x}_j|}{\zeta} \right] \left(\frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|} \right), & \text{for } |\mathbf{x}_i - \mathbf{x}_j| \leq R_i + R_j; \end{cases} \quad (20)$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ denote the position of the i th and j th particle, R_i and R_j are the radius of two particles, respectively, c_F is the collision force scale, ζ is the threshold gap at which the model is activated, ϵ_p is the stiffness parameter for collisions and E_p is another stiffness parameter smaller than ϵ_p .

Similarly, the repulsive force between a particle and a wall is given as follows:

$$F_i^W = \begin{cases} 0, & |\mathbf{x}_i - \mathbf{x}_j| > 2R_i + \zeta, \\ \frac{c_F}{\epsilon_W} \left(\frac{|\mathbf{x}_i - \mathbf{x}_j| - 2R_i - \zeta}{\zeta} \right)^2 \left(\frac{\mathbf{x}_i - \mathbf{x}_j}{|\mathbf{x}_i - \mathbf{x}_j|} \right), & |\mathbf{x}_i - \mathbf{x}_j| \leq 2R_i + \zeta, \end{cases} \quad (21)$$

where $\mathbf{x}_j$ is the position of a fictitious particle j which is located at a mirror position at the other side of the wall, and ϵ_W is another stiffness parameter.

3.5. Refilling scheme

When submerged particles move according to Eqs. (2)–(4), some of the previously solid nodes can be uncovered to become fluid nodes. Because the fluid flow equations are not solved inside of the particles, these newly introduced fluid nodes do not have their own distribution functions. According to Lallemand and Luo [37], nearby fluid nodes can be used to refill these newly introduced fluid nodes. While several

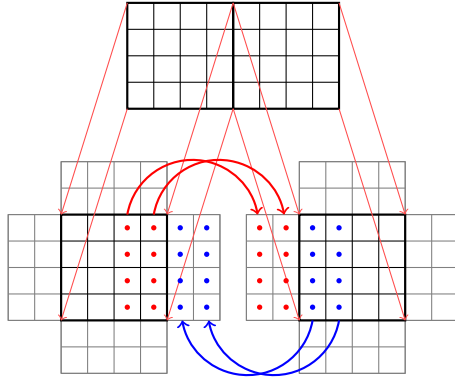


Fig. 2. Illustration of the domain decomposition technique applied to a two-dimensional configuration. The upper mesh, consisting of 8×4 cells, is divided into two subdomains of size 4×4 . Each subdomain is extended by including two layers of ghost cells in both horizontal and vertical directions. The addition of two layers of ghost cells is necessary to accurately compute the interpolated bounce-back condition on a solid surface. The arrows indicate the transfer of data from neighboring subdomains into the ghost cells, and similar data exchange occurs in other directions.

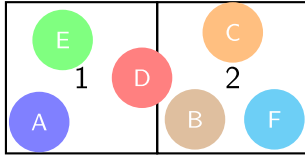


Fig. 3. An illustration of how particles are handled in domain decomposition. Given the finite-size geometry of the particles, each thread must monitor particles that are in proximity to the current subdomain.

approaches exists for such refilling, in this work, relatively simple averaged extrapolation refilling scheme was used, in which newly introduced fluid node $\mathbf{x}_{new}$ contains averaged extrapolated data from surrounding fluid nodes:

$$f_a(\mathbf{x}_{new}) = \frac{1}{N_F} \sum_{\beta} (2f_a(\mathbf{x}_{new} + \Delta t \mathbf{e}_{\beta}) - f_a(\mathbf{x}_{new} + 2\Delta t \mathbf{e}_{\beta})), \quad (22)$$

where for simplicity, time dependence of f is not shown, and N_F equals to the number of neighboring fluid nodes, while β indicates directions of these neighboring fluid nodes. The same refilling procedure is applied for energy distribution function g .

3.6. Parallel implementation

The numerical method was implemented in the Fortran 2008 programming language, while a three-dimensional domain decomposition technique was used for parallelization purposes using the Message Passing Interface (MPI) library [38]. Cartesian topology was employed to subdivide the overall computational domain into subdomains using MPI_Cart_* subroutines. Data synchronization between subdomains is achieved through exchange via ghost cells, and for the purpose of interpolated bounce-back, two layers of ghost cells were introduced (Fig. 2). This approach naturally allocates equal work to each MPI thread in the case of equally sized subdomains, rendering specific balancing strategies unnecessary for small to medium-scale applications.

A similar approach is employed for parallelization in the solid phase, where each MPI thread manages two lists of particles. The first list comprises particles whose centers belong to the current subdomain, while the second list contains information about particles close to the current subdomain that may interact with it. An example of particle arrangement is illustrated in Fig. 3, where the computational domain is divided into two subdomains, and each MPI thread maintains two lists of particles: its own list and the list of neighboring particles. For

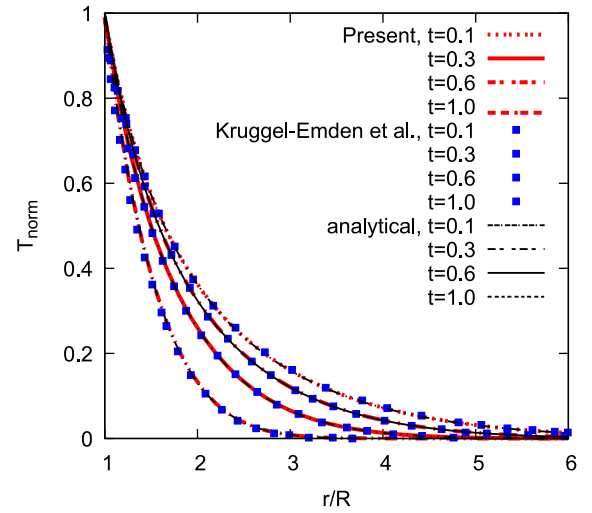


Fig. 4. Evolution of the normalized radial temperature distribution around the sphere for the case $d_p=15$. Results for Kruggel-Emden et al. correspond to the data taken from [25], and “analytical” corresponds to results for Eq. (23).

instance, in the case of the first thread, its own-particle list consists of A, E, D, while the neighbor-particle list includes B, C. Similarly, for the second thread, the own-particle list comprises B, C, F, while the neighbor-particle list contains D. At each successive time step, MPI threads exchange particle information due to their movement with the nearest neighboring threads, thereby eliminating the need for global synchronization of particle data.

The own-neighbor particle approach has been successfully applied to particle-laden cases in previous studies [39]. However, the presence of a moving solid phase may introduce work imbalances due to the accumulation of solid particles in certain spatial regions, while other regions may be devoid of particles. To simplify the code structure, we assumed that particles are uniformly distributed throughout the simulation, and no load balancing techniques were applied. However, we anticipate that this will change in future revisions of our code to address problems with a larger number of particles in greater detail.

4. Validation cases

To validate the present implementation of the Lattice-Boltzmann method with interpolated bounce-back treatment of curved boundaries, several benchmark cases were considered.

4.1. Heat diffusion of a fixed particle

Heat diffusion process around hot sphere submerged in a cold fluid is considered as the second validation case. In this case, fluid is at rest, gravity is set to zero and all boundaries are periodic, the sphere located in the center of the domain has constant temperature T_s , while initial fluid temperature is T_f . Such an unbounded domain is modeled by cubic domain of size $8d_p$, where d_p is the particle diameter. The applied time step is $\Delta t = 10^{-4}$ s, $\alpha = 10^{-6}$ m²/s and $d_p = 10^{-3}$ m. Numerically, the sphere diameter was resolved using 16 cells ($d_p^* = 15\Delta x$) and the whole domain was resolved using a 120^3 uniform mesh.

The results from numerical simulation are shown in Fig. 4, and are compared with analytical solution:

$$\frac{T(r, t) - T_f}{T_s - T_f} = \frac{R}{r} \left[1 - \operatorname{erf} \left(\frac{r - R}{\sqrt{4\alpha t}} \right) \right], \quad (23)$$

as well as numerical simulations from the literature [25]. Our implementation shows very good results, indicating proper implementation of the thermal boundary conditions.

Table 1
Parameters used in the experiment performed by ten Cate et al. [40].

Case	I	II	III	IV
μ , g/cm s,	3.73	2.12	1.13	0.58
ρ , g/cm ³ ,	0.97	0.965	0.962	0.96

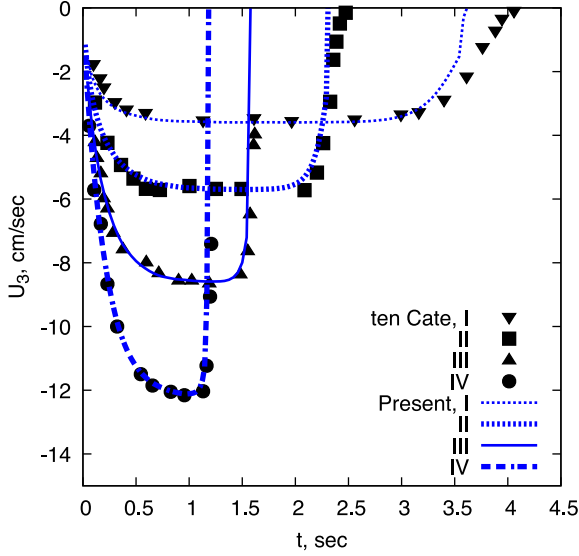


Fig. 5. Evolution of the vertical velocity of the settling particles compared to the results from [40]. Cases I-IV correspond to four different parameter sets as shown in Table 1.

4.2. Settling of a single isothermal particle

In the next test case, coupling of the dynamics of the fluid and solid phases is validated by studying the settling of single spherical particle in a viscous fluid. This case was studied experimentally by ten Cate [40].

Four different combinations of the fluid viscosity and density are considered as indicated in Table 1. The enclosure size is $10 \times 10 \times 16$ cm, spherical particle diameter is $D = 1.5$ cm with density of $\rho_s = 1.12$ g/cm³. Fig. 5 present results from numerical simulation of this validation case. The evolution of the settling velocity is in good agreement with the experimental results.

4.3. Three-dimensional Rayleigh-Bénard convection

Thermal flow dynamics was studied in three-dimensional configurations for Rayleigh-Bénard (RB) convection. The cell is heated from the bottom wall and cooled at the top wall, and the side walls are adiabatic. Non-zero gravitational acceleration and negative vertical temperature gradient leads to the heat transfer from bottom to top wall. The RB convection is mainly governed by the Rayleigh number (Ra) and Prandtl number (Pr) defined as:

$$Ra = \frac{\beta \Delta T g H^3}{\nu \alpha}, \quad Pr = \frac{\nu}{\alpha}. \quad (24)$$

For a single-phase RB convection, the average Nusselt number can be computed in three different manners:

$$\begin{aligned} Nu &= -\frac{H}{\Delta T} \left\langle \left(\frac{\partial T}{\partial z} \right)_{z=0} \right\rangle_S = -\frac{H}{\Delta T} \left\langle \left(\frac{\partial T}{\partial z} \right)_{z=H} \right\rangle_S \\ &= 1 + \frac{\langle u_z T \rangle_V}{\alpha \Delta T / H}, \end{aligned} \quad (25)$$

where $\langle (\cdot) \rangle_S$ and $\langle (\cdot) \rangle_V$ indicate surface or volume average, respectively, further time average is recommended for unsteady flows. The last form emphasizes the combined role of convective heat transport and diffusive heat transport.

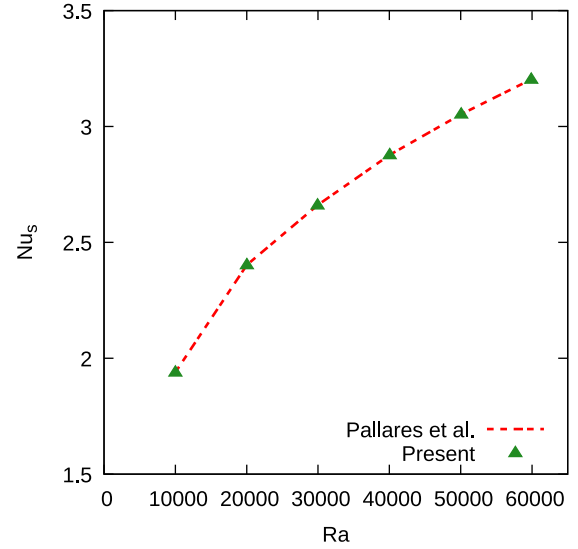


Fig. 6. The Nusselt number as a function of the Rayleigh number for the case of three-dimensional Rayleigh-Bénard convection. The results for Pallares et al. correspond to the data from [41].

Table 2

The Nusselt number in single-phase Rayleigh-Bénard convection obtained at three different grid resolutions, with Rayleigh number $Ra = 10^7$ and Prandtl number $Pr = 0.71$. The reported results of the Nusselt number for the same setting from the literature is in the range of 16.1 to 16.3 [42].

Mesh	Nu_{top}	Nu_{bottom}	Nu_{bulk}	t_{avg}
160^3	16.24 ± 0.08	16.24 ± 0.08	16.09 ± 0.13	950
256^3	16.24 ± 0.08	16.37 ± 0.07	16.26 ± 0.17	450
320^3	16.04 ± 0.11	16.08 ± 0.12	16.07 ± 0.19	200

The uniform numerical mesh of the cubic domain with a resolution of 64^3 is used first. (Fig. 6) shows the comparison between the Nusselt number obtained from simulations and those in the literature [41], and a good agreement of Nu - Ra dependence can be observed.

To further validate our implementation, the grid convergence test is performed for the RB convection at $Ra = 10^7$. The Nusselt number obtained based on temperature gradients at the bottom wall, those at the top wall, and the volume averages of convective flux at different resolutions is shown in Table 2. The Nusselt numbers obtained at the statistically stationary state with different resolutions are close to each other. The reported results of the Nusselt number for the same setting from the literature are in the range of 16.1 to 16.3 [42]. The relative error between our results and that in the literature is about 1%.

4.4. Settling of a single thermal particle

To verify the ability of the lattice Boltzmann method to handle curved thermal surfaces, simulations are performed for both the single thermal particle settling problem and the multi thermal particle settling problem. In the first case, a single sphere with diameter d_p is placed in a domain. By varying the density ratio $\rho_r \equiv \rho_p / \rho_f$ between the particle and fluid, different terminal settling velocities can be obtained. The density ratio is adjusted according to the given reference Reynolds number,

$$Re_{ref} = \frac{d_p U_{ref}}{\nu}, \quad (26)$$

where the theoretical sedimentation velocity U_{ref} in the vertically unbounded domain is calculated by

$$U_{ref} = \sqrt{\frac{4d_p g}{3}(\rho_r - 1)}. \quad (27)$$

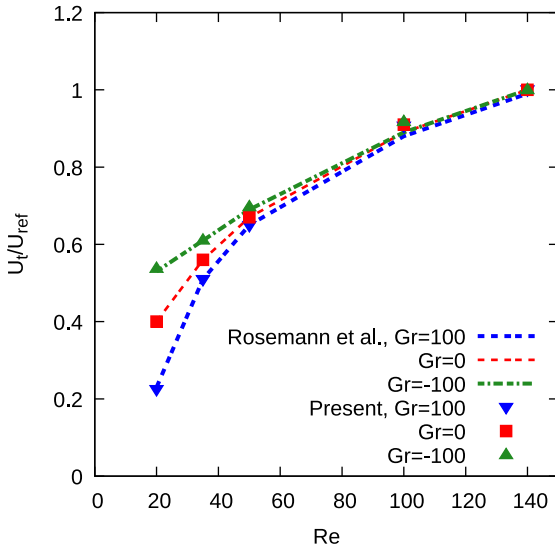


Fig. 7. Settling velocity of one particle for varying Grashof and Reynolds numbers compared to the results from [26].

Table 3

The properties of the six isothermal spherical particles with different densities and temperatures but equal size at initial stage.

Particle no.	$r_p(t_0)[d_p]$	$T_p[K]$	Re_{ref}
1	(1.5,1.3,14)	320	50
2	(1.7,2.0,17)	300	50
3	(1.5,3.3,10)	300	35
4	(3.3,1.7,11)	320	35
5	(1.5,3.5,12)	320	35
6	(3.1,1.7,13)	300	35

Additionally, the settling velocities are examined for different Grashof numbers. The Grashof number, denoted as Gr , is calculated as

$$Gr = \frac{g\beta d_p^3(T_p - T_{ref})}{\nu^2}, \quad (28)$$

where T_p represents the particle's temperature, and T_{ref} is the wall temperature. Three Grashof numbers $Gr = (-100, 0, 100)$ are considered here, and Fig. 7 presents the results for the terminal velocity normalized by the reference velocity U_{ref} . The results agree well with those in [26].

4.5. Settling of multiple thermal particles

To further examine the capabilities of the present algorithm, simulations were conducted for the multi-particle settling problem. Six spherical particles of equal diameter but varying densities and temperatures are placed inside an enclosure with dimensions of $5d_p \times 5d_p \times 20d_p$. The particles are positioned close to each other but are not allowed to collide. Table 3 provides the properties of the six isothermal spherical particles involved in the simulation, and the density of the particles is determined according to Eq. (26)–(27). Table 4 lists the resulted Nusselt number and drag coefficient of each particle at time $t = 3.24$ s, which are compared with the results in [26], and the agreement is good.

4.6. Parallel performance

Finally, efficient interface-resolved simulations of particle-laden flows require the application of parallelization techniques to speed up computations on modern multi-core/multi-CPU computers. To evaluate the effectiveness of our approach in efficiently conducting fine mesh

Table 4

Nusselt numbers and drag coefficients for the time $t = 3.24$ s compared to the results from [26].

Particle no.	Nu [26]	Nu	Cd [26]	Cd
1	5.42	5.22	0.99	0.99
2	6.01	6.17	0.99	0.99
3	4.43	4.49	1.01	1.01
4	4.51	4.46	1.01	1.01
5	5.42	5.39	0.98	0.98
6	5.60	5.65	0.96	0.96

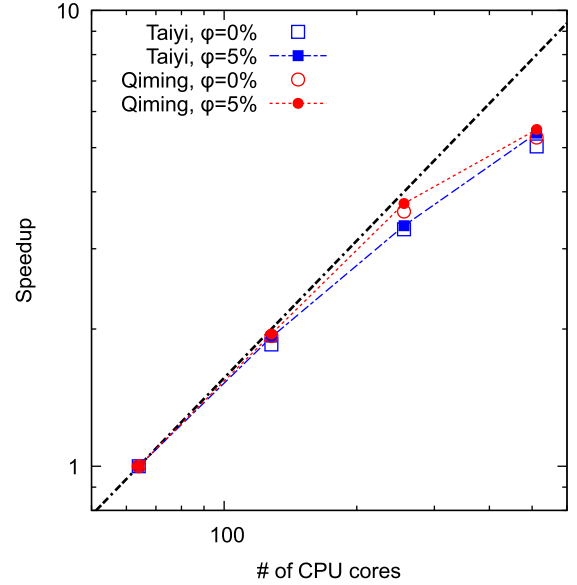


Fig. 8. The strong scaling speed-up benchmark results were conducted on the Taiyi and Qiming computer clusters, a fixed grid resolution of $N_x \times N_y \times N_z = 384 \times 384 \times 384$ was adopted.

simulations, we present the results from the strong scaling benchmark in Fig. 8, obtained from two clusters (Qiming and Taiyi) at our institution. The results for both single-phase ($\phi = 0\%$) and particle-laden ($\phi = 5\%$) cases are depicted. Each node of the Taiyi computer cluster is equipped with two Xeon Gold 6148 CPUs (2.4 GHz), whereas each node of the Qiming cluster consists of two Intel Xeon E5-2690v3 CPUs (2.6 GHz). Our Fortran-MPI implementation has average parallel efficiency of 87% and 66% on 256 and 512 CPU cores respectively. Further analysis of the runtime breakdown for the particle-laden case revealed that activating communication subroutines for the solid-phase resulted in less than 1% increase in total runtime compared to the single-phase case. This suggests that the contribution from tracking the particles does not lead to excessive communication costs. However, due to the application of boundary treatment using the interpolated bounce-back method, the overall runtime increased by an average of 16%.

5. Particle-laden thermal convection

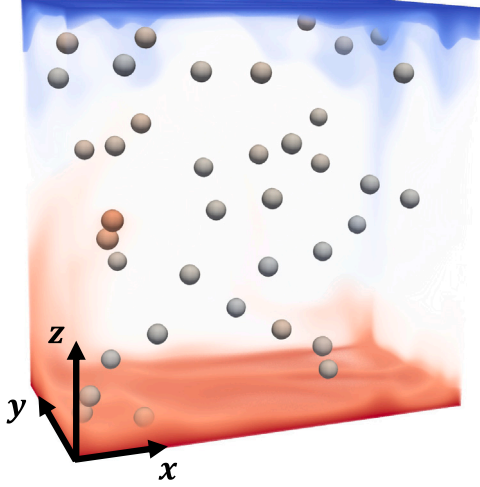
Now we consider the Rayleigh-Bénard convection laden with finite-size spherical particles. The three-dimensional RB system is heated (T_h) from the bottom and cooled (T_c) from the top at constant temperatures with the height being H , and the remaining four solid walls are adiabatic.

The characteristic scales used for the non-dimensionalization are the length scale (height of the domain, H), and the velocity scale ($\sqrt{g\beta\Delta TH}$). The temperature is made dimensionless as $T_f^* = (T - T_0)/\Delta T$, where $T_0 = (T_c + T_h)/2$.

Table 5

Physical and numerical parameters adopted in the particle-laden Rayleigh-Bénard convection.

Ra	Pr	d_p^*	ϕ	$N_x \times N_y \times N_z$
10^7	0.71	1/20	(0, 1%, 3%, 5%, 7%, 9%)	$320 \times 160 \times 320$

**Fig. 9.** The instantaneous temperature field for the particle-laden Rayleigh-Bénard convection with volume fraction $\phi = 1\%$, red color corresponds to hot thermal plume, blue color corresponds to cold thermal plume, and particles are colored based on their temperature ($T \in (0.5, 1)$).

In this work, we assume that the particle density ρ_p is equal to the fluid density ρ_f . Under this condition, the system contains 7 independent governing dimensionless parameters: Ra , Pr , $d_p^* = d_p/H$, $c_{p,p}/c_{p,f}$, ϕ , L/H , and W/H , where ϕ is the particle volume fraction, L , W and H are the domain length, width and height. The two aspect ratios are fixed to $L/H = 1$ and $W/H = 0.5$.

In this section, we will explore the effect of particle volume fraction on the heat transfer efficiency in the system, and six different volume fractions ($\phi = 0\%, 1\%, 3\%, 5\%, 7\%, 9\%$) are considered. The parameters used are shown in Table 5, the Rayleigh number is fixed at $Ra = 10^7$, the Prandtl number is fixed at $Pr = 0.71$, the ratio of the heat capacity between the particle and the fluid ($c_{p,p}/c_{p,f}$) is 10, and the dimensionless particle diameter is $d_p^* = 1/20$.

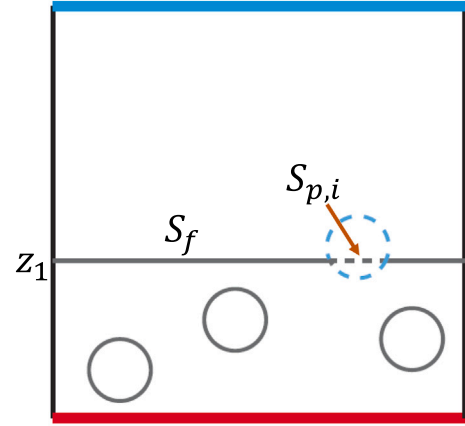
Fig. 9 shows the instantaneous temperature field for the particle-laden Rayleigh-Bénard convection with particle volume fraction 1%. As shown in the figure, the thermal plumes generated from the thermal boundaries travel along the vertical direction near the side walls, enhancing the thermal mixing and heat transfer. In addition, the presence of the solid particles would introduce other mechanisms to modulate the heat transfer. As can be seen in Fig. 9, the hot and cold particles co-exist in the system, and their large-scale-circulation modulates the heat transfer of the system.

5.1. The heat transfer efficiency in the system

One of the important problems in the Rayleigh Bénard convection is the heat transfer efficiency, which is usually characterized by the Nusselt number.

As shown in Fig. 10, for the particle-laden RB convection, the averaged heat flux across a horizontal plane at the vertical location z_1 can be expressed as:

$$\frac{1}{S_{bot}} \left[\int_{S_f} \left(u_z T_f - \alpha \frac{\partial T_f}{\partial z} ds_f \right) + \frac{c_{p,p}}{c_{p,f}} \sum_i u_{p,z} T_p S_{p,i} \right], \quad (29)$$

**Fig. 10.** A sketch for the calculation of Nusselt number in the particle-laden thermal flows.**Table 6**The obtained Nusselt numbers for different cases in particle-laden Rayleigh-Bénard convection with $Ra = 10^7$, $Pr = 0.71$, the grid resolution $N_x \times N_y \times N_z = 320 \times 160 \times 320$. The statistics are averaged over 300 dimensionless time.

ϕ	Nu_{top}	Nu_{bottom}	Nu_{bulk}	Nu_{fluid}
0%	16.20 ± 0.13	16.21 ± 0.14	16.24 ± 0.27	16.24 ± 0.27
1%	16.50 ± 0.11	16.46 ± 0.15	16.67 ± 0.71	15.42 ± 0.33
3%	16.61 ± 0.13	16.56 ± 0.12	16.82 ± 1.27	13.44 ± 0.34
5%	16.98 ± 0.13	16.98 ± 0.11	17.12 ± 1.31	12.05 ± 0.33
7%	17.23 ± 0.12	17.20 ± 0.14	17.47 ± 1.14	10.63 ± 0.49
9%	17.55 ± 0.11	17.49 ± 0.11	17.61 ± 1.19	9.52 ± 0.34

where S_{bot} is the surface area of the bottom wall, S_f is the plane area cutting through the fluid domain, and $S_{p,i}$ is the plane area cutting through the i th particle.

Integrate Eq. (29) over the vertical direction z from the bottom to top, and assume that the temperature gradient $\frac{\partial T_f}{\partial z}$ is continuously defined. The following expressions for the overall Nusselt number for the particle-laden thermal flow is obtained:

$$Nu = (1 - \phi) \left[1 + \frac{\langle u_z T \rangle_{FV}}{\alpha \Delta T / H} \right] + \phi \frac{c_{p,p}}{c_{p,f}} \frac{\langle u_{p,z} T_p \rangle_P}{\alpha \Delta T / H} \equiv Nu_{fluid} + Nu_{particle}. \quad (30)$$

The first term on the right-hand-side (RHS) represents the heat flux transferred by the bulk fluid itself, and the second term on the RHS means the contribution to heat flux from the solid particles. This extends the expression of Eq. (25) for single phase flow, which includes the contribution from the particles to the heat transfer efficiency in the system. It should be noted that the Nu of the particle-laden RB can also be calculated by the ratio between the heat flux at the bottom/top wall and the conductive heat flux, which is identical with the first two methods in Eq. (25).

The overall heat transfer rate in terms of the Nu number averaged over a period of $t_{avg} = 300H/\sqrt{g\beta\Delta TH}$ is displayed in Fig. 11, and the corresponding data are listed in Table 6. The Nusselt number in Table 6 takes the form of $\bar{Nu} \pm \sigma_{\bar{Nu}}$, where $\sigma_{\bar{Nu}}$ is the uncertainty for each case to demonstrate the accuracy of the data. The definition of the uncertainty can be found in Eq. (8) in the work of Wang et al. [43]. The overall heat transfer rate monotonically increases with the increase in the volume fraction of particles, and the largest enhancement is about 8.43% in the case with a 9% particle volume fraction. Demou et al. [44] reported the similar phenomenon, and they believed that the presence of the solid phase enhanced the plume in the bulk region, leading to the increase of heat transfer efficiency. Using Eq. (30), we can separate the heat transfer contributed from the carrier fluid and the particles, and the results are reported in Table 6 and Figs. 11(b) and 11(c). As shown in the last column of Table 6, the heat flux transported by the fluid is

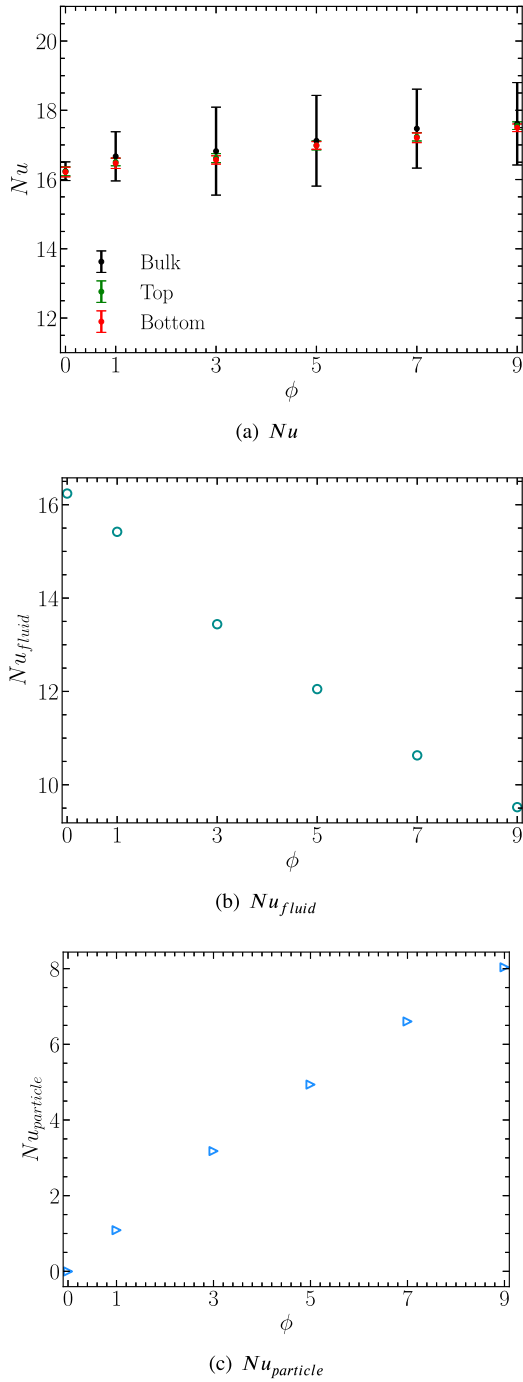


Fig. 11. (a) The obtained Nusselt numbers for different cases in particle-laden Rayleigh-Bénard convection using three methods. The contribution of Nusselt number from (b) the bulk fluid and (c) the moving particles.

attenuated due to the presence of solid particles, but the particle serves as additional heat ‘carriers’, enhancing the heat transfer efficiency in the system (see also in Fig. 11(c)).

5.2. Modulation to the mean profiles

As the heat transfer is closely related to the fluid velocity and temperature, we shall study the modulation of the fluid velocity and temperature fields by the solid particles. Fig. 12 shows the wall-normal temperature profiles which are averaged over the fluid domain for

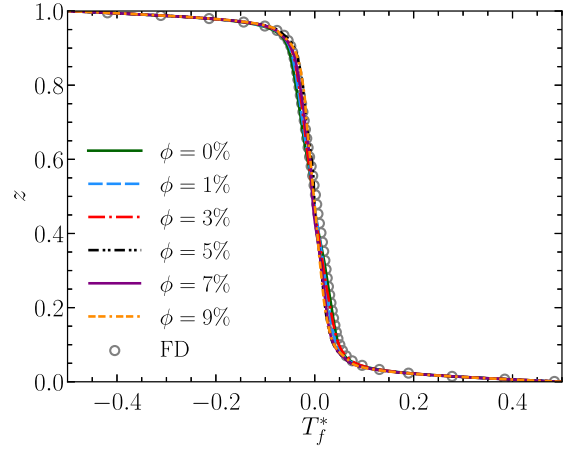


Fig. 12. The mean flow temperature profiles. The data are averaged over 200 dimensionless time. The gray circles correspond to the result from the finite different method with the same Ra and Pr in single-phase RB convection.

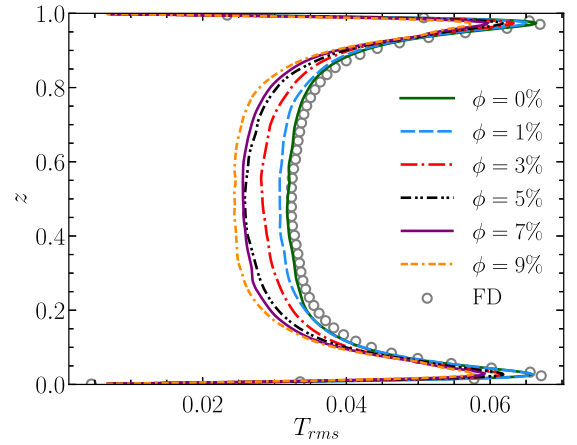


Fig. 13. The profiles of r.m.s fluid temperature as a function of the vertical location. The data are averaged over 200 dimensionless time.

different cases, negligible difference can be observed for different particle volume fractions, and all profiles exhibit a sharp gradient near the solid wall. As the volume fraction in this study is relatively small, the modulation to the temperature profile by different particle volume fractions is similar, which is also observed in Demou et al. [44], in which the temperature profiles were similar for different cases up to $\phi=30\%$. The gray circle represents the temperature profile of the single-phase RB convection with the same configuration obtained from the finite-difference method known as AFID for comparison. This open-source finite-difference code developed by van der Poel et al. [45] is adopted, and the non-uniform grids with $N_x \times N_y \times N_z = 162 \times 96 \times 162$ is used in the finite-difference simulation, with 162 stretched grids in the vertical direction, while the LBM in this study uses $N_x \times N_y \times N_z = 320 \times 160 \times 320$ uniform grids. From Fig. 12, the agreement between the LBM and the finite-difference method is good. In addition, the Nusselt numbers obtained from our simulations and finite-difference simulation are 16.24 and 16.38, respectively, which are comparable.

While the particle volume fraction has a negligible impact on the mean temperature profile of the carrier flow, the suppression on the root-mean-square of the temperature (Fig. 13) is enhanced with the increase of particle volume fraction. The gray circle is the r.m.s temperature profile (Fig. 13) of the single-phase RB convection with the same configuration obtained using the finite-difference method, the small

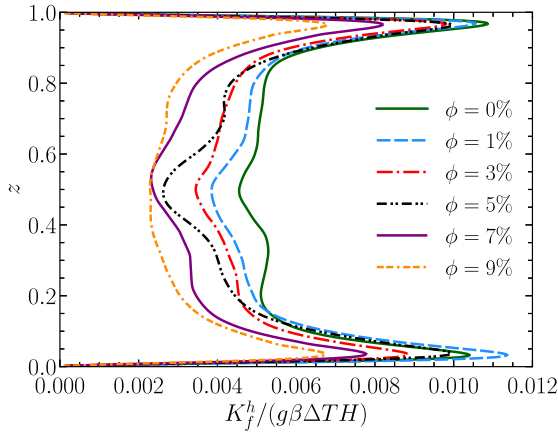


Fig. 14. The profiles of horizontal r.m.s fluid velocity fluctuations. The data are averaged over 200 dimensionless time.

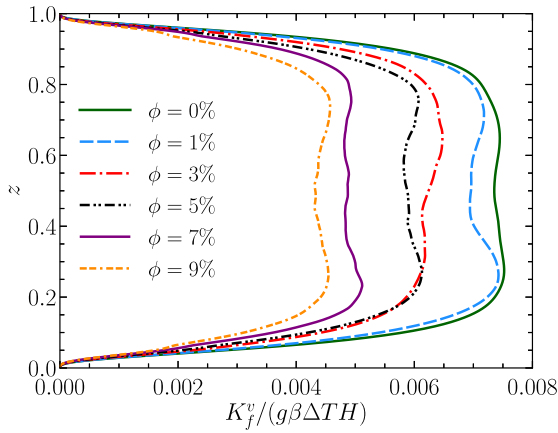


Fig. 15. The profiles of vertical r.m.s fluid velocity fluctuations. The data are averaged over 200 dimensionless time.

discrepancy might be the reason of the lower resolution used in finite-difference simulation. The moderate attenuation of the temperature fluctuations with increasing particle volume fraction is consistent with the reduced rms velocity fluctuations, to be presented below.

The presence of the solid particles would also modulate the velocity field, and the modulations to the horizontal and vertical components of the kinetic energy are investigated here. The horizontal and vertical contributions to the kinetic energy are defined as:

$$\begin{aligned} K_f^h &= \frac{1}{2} [(u_f, rms)^2 + (v_f, rms)^2], \\ K_f^v &= \frac{1}{2} (w_f, rms)^2. \end{aligned} \quad (31)$$

Figs. 14 and 15 show the vertical profiles of K_f^h and K_f^v , and it is observed that the presence of solid particles always suppress the turbulent kinetic energy, indicating the weakening of large-scale circulation. The attenuation of kinetic energy in turbulent flow is commonly reported in finite-size particle-laden turbulent flows, both for isothermal and thermal flows [46]. In addition, the attenuation of K_f^h and K_f^v increases with increasing volume fraction due to the increased effective viscosity.

From Eq. (30), we know that the Nu number contributed from the fluid domain is $(1 - \phi) \left[\frac{1 + \langle u_z T' \rangle_{FV}}{\alpha \Delta T / H} \right]$. As the mean vertical velocity is zero, $\langle u_z T \rangle_{FV}$ can be written as: $\langle u_z T' \rangle_{FV} + \langle u_z \bar{T} \rangle_{FV}$, where the superscript (') indicates the fluid temperature and velocity fluctuations, and $\bar{T}$ is the average temperature in the fluid domain. While the profiles of the r.m.s

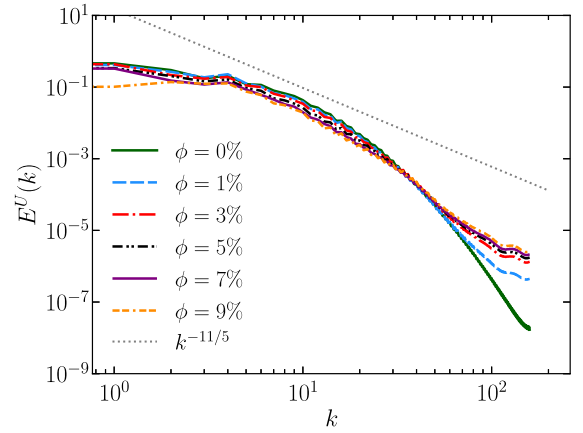


Fig. 16. The spectra of the turbulent kinetic energy for different cases. While the large-scale energy is suppressed by the particles, the presence of particles typically enhance the small-scale fluctuations.

temperature and turbulent kinetic energy are suppressed by the solid particles, the heat transported by the fluid would be smaller, which is consistent with the attenuation of the Nusselt number contributed from the fluid domain. In addition, the tendency of the monotonic decrease of the r.m.s temperature and turbulent kinetic energy when varying the particle volume fraction is consistent with the attenuation of the heat flux transported by the fluid.

5.3. Modulation of the energy spectra

In addition to the mean statistics of the flow field, the modulation to the different scales of the bulk flow is important. To investigate the turbulence modulation at different scales due to the particles, the spectra of the turbulent kinetic energy (TKE) and temperature fluctuations are studied. As the fluid is confined by the solid walls (no-slip walls) in three directions, the three-dimensional discrete sine transformation is performed to obtain the spectrum of the turbulent kinetic energy. In addition, as the four side walls are adiabatic, the two-dimensional discrete cosine transformation is applied in the horizontal plane, and the average over the vertical direction is carried out afterwards to obtain the spectrum of the temperature. In computing the spectra, the velocity inside the particles are assumed to take the form of rigid body motion of the particle, and the temperature inside the particles are set to the particle temperature. It should be noted that the spectra shown are further averaged over time.

As shown in Fig. 16, the turbulent kinetic energy (TKE) is attenuated at large scale and augmented at the small scale, which is usually observed in finite-size particle-laden isothermal turbulence. The pivot wavenumber dividing the two regions is close to the scale of the particle size. The small-scale fluctuation is usually enhanced due the unsteady wakes generated behind the particles. It can also be observed that the enhancement of small-scale fluctuation increases with increasing particle volume fraction, which might be the reason for larger slip velocity between the two phases.

Slightly different from the energy spectrum of the turbulent kinetic energy, the temperature field (shown in Fig. 17) is only attenuated at the largest scale, while the intermediate wavenumber seems to be unchanged with the presence of solid particles. Similar to the TKE spectrum, the temperature is also augmented at the small scales. The enhanced small-scale fluctuations of temperature would lead to a higher probability of the generation of hot/cold plumes, which will enhance the heat transfer efficiency. The energy spectra illustrate the competing mechanisms brought by the particles: enhanced heat transfer at the scale of particle size due to the local disturbance flows around the particles, and reduced large-scale turbulence fluctuations due to the

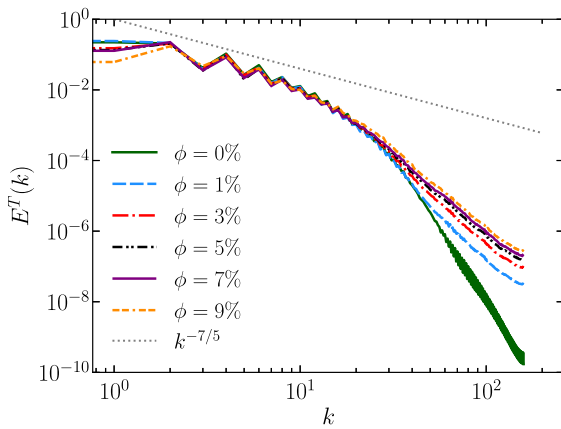


Fig. 17. The spectra of the temperature for different cases. Similar to the TKE spectrum, the particles attenuate the large-scale energy and augment the small-scale energy.

additional viscous dissipation near the particle surfaces. But more quantitative studies should be performed in our future work to demonstrate the two-way interactions between the finite-size solid particles and the thermal turbulence.

5.4. Thermal plumes

As we know, the most important coherent structure in turbulent thermal convection is the thermal plumes, and they are responsible for coherent heat transfer in the system. To demonstrate the influence of particles on the thermal plumes at different scales, the thermal plumes for different cases are extracted and are shown in Fig. 18. The approach of van der Poel et al. [47] is adopted to detect the thermal plumes, the thermal plumes in the bulk is defined as a region where:

$$|T(x, y, z, t) - \langle T \rangle_{x,y,t}| > cT_{rms} \quad \text{and}, \quad \sqrt{PrRa}|\hat{u}_z(x, y, z, t)\hat{T}(x, y, z, t)| > cNu, \quad (32)$$

where $\hat{u}_z$ is the vertical velocity normalized by $\sqrt{g\beta\Delta TH}$, $\hat{T}$ is the temperature normalized by $(T_h + T_c)/2$, and the empirical constant c is chosen to be 1.2 [47]. In extracting the thermal plumes, the temperature inside the particle domain is defined by particle temperature, such that the resulting thermal plumes can be regarded as the combination of thermal plumes of the fluid phase and those of the solid phase.

Fig. 18(a) shows the thermal plumes extracted from the single-phase RB convection, and the majority is the large-scale thermal plumes. With the increase of particle volume fraction (Fig. 18(b)~Fig. 18(c)), the large-scale thermal plumes are weakened, and more small-scale plumes are generated, as the enhanced small-scale fluctuations of temperature would lead to a higher probability of the generation of small-scale hot/cold plumes. Meanwhile, the suppression of the large-scale thermal plumes becomes larger as the particle volume fraction increases, which would result in the smaller heat transfer efficient from the thermal fluid itself. Thus, we believe that the overall heat transfer enhancement in the system is mainly due to the additional heat flux transported by the particles.

6. Summary and conclusions

In this work, a robust and efficient simulation method for particle-laden Rayleigh-Bénard convection is developed. The present work utilizes the double distribution function-based thermal lattice-Boltzmann method (TLBM), which enables successful simulations of multiphase fluid-thermal interactions. The no-slip boundary of the moving solid particles is handled by the interpolated bounce-back scheme. Additionally, Galilean invariant momentum exchange and heat exchange

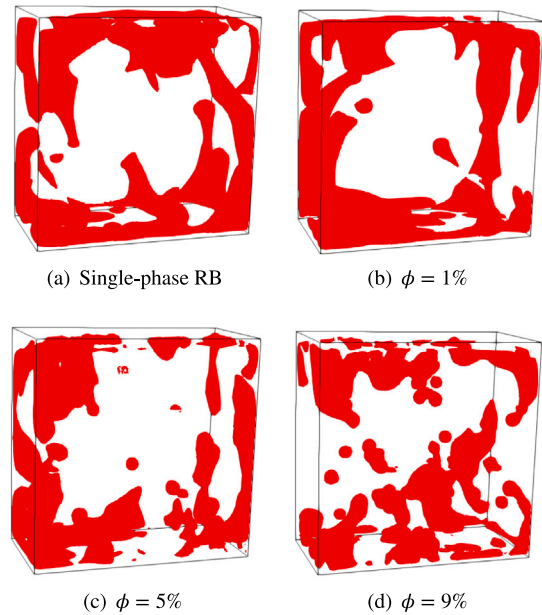


Fig. 18. Thermal plumes extracted from flows with different particle volume fractions. As the particle volume fraction increases, the larger-scale plumes are destroyed, and more small-scale thermal plumes are generated.

approaches are employed for hydrodynamic force and heat transfer calculation at the solid boundaries.

The accuracy of the current method is verified using five benchmark cases, including single-particle settling at isothermal conditions, three-dimensional Rayleigh-Bénard convection, heat diffusion around stationary solid sphere and settling problem for single and six particles at different temperatures relative to ambient fluid temperature. For all benchmark cases, very good agreements are found with experimental and numerical data from the literature, indicating the correctness of the applied model and of its implementation.

Furthermore, we explore the modulation of Rayleigh-Bénard flows by the presence of freely moving finite-size particles with different volume fractions. A modified flow-based Nusselt number evaluation is proposed for the particle-laden thermal system. The heat transfer efficiency monotonically increases with the increasing particle volume fraction in our study while the particle volume fraction is of small to moderate values. Although the Nusselt number contributed from the fluid is attenuated by the particle, the finite-size particles also contribute to the heat transfer process, enhancing the overall heat transfer efficiency. Besides the heat transfer efficiency enhancement, the particles also enhance small-scale turbulence and temperature fluctuations, which would increase the probability of the generation of small-scale thermal plumes. These preliminary results indicate that the finite-size particle can also be used to enhance the heat transfer efficiency in the thermal convection system, which may have many potential applications in designing industrial facilities.

CRediT authorship contribution statement

Hongcheng Wu: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Kairzhan Karzhaubayev:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Jie Shen:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Lian-Ping Wang:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Link between LBM and macroscopic governing equations

The classical lattice Boltzmann method is an incompressible Navier–Stokes solver. The macroscopic Navier–Stokes equation can be recovered from the mesoscopic discrete lattice Boltzmann equation using the Chapman–Enskog expansion. Firstly, Taylor expand the left-hand side (LHS) terms of the lattice Boltzmann equation (Eq. (5)) to the order of $\mathcal{O}(\Delta t^2)$:

$$\begin{aligned} & \Delta t (\partial_t + e_{\alpha,j} \partial_j) f_\alpha + \frac{(\Delta t)^2}{2} (\partial_t + e_{\alpha,j} \partial_j)^2 f_\alpha \\ &= -\frac{1}{\tau_f} [f_\alpha - f_\alpha^{eq}(\mathbf{x}, t)] + \Delta t F_\alpha, \\ & \Delta t (\partial_t + e_{\alpha,j} \partial_j) g_\alpha + \frac{(\Delta t)^2}{2} (\partial_t + e_{\alpha,j} \partial_j)^2 g_\alpha \\ &= -\frac{1}{\tau_g} [g_\alpha - g_\alpha^{eq}(\mathbf{x}, t)]. \end{aligned} \quad (33)$$

Use the multiscale-expansion:

$$\phi_\alpha = \phi_\alpha^{(0)} + \varepsilon \phi_\alpha^{(1)} + \varepsilon^2 \phi_\alpha^{(2)} + \dots, \quad (34a)$$

$$\partial_t = \varepsilon \partial_{t0} + \varepsilon^2 \partial_{t1}, \quad (34b)$$

$$\partial_j = \varepsilon \partial_{j0}, \quad (34c)$$

$$b_j = \varepsilon b_{j0}, \quad (34d)$$

where ϕ_α is the distribution function, can be used to represent either f_α or g_α . b_j is the external body force. Substitute Eq. (34) into Eq. (33), and then group each term by the small parameter ε . The following equations can be obtained. For velocity field:

$$\mathcal{O}(\varepsilon^0) : f_\alpha^{(0)} = f_\alpha^{eq}, \quad (35)$$

$$\begin{aligned} \mathcal{O}(\varepsilon^1) : & (\partial_{t0} + e_{\alpha,j} \partial_{j0}) f_\alpha^{(0)} = \\ & -\frac{1}{\tau_f} \Delta t f_\alpha^{(1)} + \left(1 - \frac{1}{2\tau_f}\right) \frac{b_{j0}(e_{\alpha,j} - u_j)}{c_s^2} f_\alpha^{eq}, \end{aligned} \quad (36)$$

$$\begin{aligned} \mathcal{O}(\varepsilon^2) : & \partial_{t1} f_\alpha^{(0)} + (\partial_{t0} + e_{\alpha,j} \partial_{j0}) f_\alpha^{(1)} \\ & + \frac{\Delta t}{2} (\partial_{t0} + e_{\alpha,j} \partial_{j0})^2 f_\alpha^{(0)} = -\frac{1}{\tau_f} \Delta t f_\alpha^{(2)}, \end{aligned} \quad (37)$$

and for temperature field:

$$\mathcal{O}(\varepsilon^0) : g_\alpha^{(0)} = g_\alpha^{eq}, \quad (38)$$

$$\mathcal{O}(\varepsilon^1) : (\partial_{t0} + e_{\alpha,j} \partial_{j0}) g_\alpha^{(0)} = -\frac{1}{\tau_g} \Delta t g_\alpha^{(1)}, \quad (39)$$

$$\mathcal{O}(\varepsilon^2) : \partial_{t1} g_\alpha^{(0)} + \left(1 - \frac{1}{2\tau_g}\right) (\partial_{t0} + e_{\alpha,j} \partial_{j0}) g_\alpha^{(1)} = -\frac{1}{\tau_g} \Delta t g_\alpha^{(2)}. \quad (40)$$

From the $\mathcal{O}(\varepsilon^0)$ terms (Eqs. (35) and (38)), it is known that:

$$\sum_i f_\alpha^{(k)} = 0, \quad \sum_i g_\alpha^{(k)} = 0, \quad \text{for } k \geq 1. \quad (41)$$

The zeroth and first order moment of $\mathcal{O}(\varepsilon^1)$ terms recovers the Euler equation:

$$\begin{aligned} \partial_{t0} \rho + \partial_{j0} (\rho u_j) &= 0, \\ \partial_{t0} \rho u_i + \partial_{j0} (\rho u_i u_j) + \partial_{j0} (p \delta_{ij}) &= \rho b_{i0}, \end{aligned} \quad (42)$$

$$\partial_{t0} T + \partial_{j0} (T u_j) = 0,$$

and the relation:

$$\sum_i e_{\alpha,i} f_\alpha^{(1)} = -\frac{\Delta t}{2} \rho b_{i0}, \quad \sum_i e_{\alpha,i} f_\alpha^{(k)} = 0 \quad \text{for } k \geq 2. \quad (43)$$

The corresponding macroscopic equations of the $\mathcal{O}(\varepsilon^2)$ terms are:

$$\partial_{t1} \rho = 0, \quad (44a)$$

$$\partial_{t1} \rho u_i = \partial_{j0} \tau_{ij}, \quad (44b)$$

$$\partial_{t1} T = \alpha \partial_{j0} (\partial_{j0} T) + \left(\tau_g - \frac{1}{2}\right) \Delta t \partial_{j0} \left(-\frac{T}{\rho} \partial_{j0} p + T b_{j0}\right), \quad (44c)$$

with $\tau_{ij} = \mu [\partial_{j0} u_i + \partial_{i0} u_j]$, $\mu = \left(\tau_f - \frac{1}{2}\right) \rho c_s^2 \Delta t$, $\nu_f = \mu/\rho = \left(\tau_f - \frac{1}{2}\right) c_s^2 \Delta t$ and $\alpha = \left(\tau_g - \frac{1}{2}\right) \Delta t c_s^2$. It should be noted that, by deriving Eq. (44b), the $\mathcal{O}(Ma^3)$ and higher-order terms are neglected. The second term on the right hand side of Eq. (44c) is also a high-order term in the convective-diffusion equation, which is usually ignored.

In summary, by summing up all the results obtained from different scales, the macroscopic equations recovered are:

$$\partial_t \rho + \partial_j (\rho u_j) = 0, \quad (45a)$$

$$\partial_t \rho u_i + \partial_j (\rho u_i u_j) = -\partial_i p + \partial_j \tau_{ij} + \rho b_i, \quad (45b)$$

$$\partial_t T + \partial_j T u_j = \alpha \partial_j (\partial_j T), \quad (45c)$$

with $\tau_{ij} = \mu [\partial_j u_i + \partial_i u_j]$.

Appendix B. The lifting relation and the boundary condition in thermal LBM

The lifting relation, which is a truncated explicit expression of the distribution function that associate the distribution function and the hydrodynamic variables, is important for designing the boundary condition and constructing the initial condition. From the Chapman–Enskog expansion, the distribution function can be expressed by the equilibrium part and non-equilibrium part:

$$f_\alpha = f_\alpha^{(eq)} + \varepsilon f_\alpha^{(1)} + \mathcal{O}(\varepsilon^2). \quad (46)$$

As $\varepsilon f_\alpha^{(1)}$ term is adequate to recover the Navier–Stokes equations, the explicit form of $\varepsilon f_\alpha^{(1)}$ term will be derived in the following. From the $\mathcal{O}(\varepsilon^1)$ term, the non-equilibrium distribution of f_α is related to the temporal and spatial derivatives of the equilibrium distribution, which can be obtained using the Euler equation:

$$\begin{aligned} \varepsilon f_\alpha^{(1)} &= -\tau_f \Delta t (\partial_t + e_{\alpha,j} \partial_j) f_\alpha^{(0)} + \left(\tau_f - \frac{1}{2}\right) \Delta t \frac{b_j (e_{\alpha,j} - u_j)}{c_s^2} f_\alpha^{eq} \\ &= -\tau_f \Delta t \omega_\alpha \left[\frac{Q_{\alpha,km}}{c_s^2} \rho \partial_k (u_m) - \frac{e_{\alpha,k}}{c_s^2} \partial_j (\rho u_j u_k) \right. \\ &\quad \left. + e_{\alpha,j} \frac{Q_{\alpha,km}}{2c_s^4} \partial_j (\rho u_k u_m) - \frac{Q_{\alpha,km}}{2c_s^4} \partial_j (\rho u_m u_k u_j) \right] \\ &\quad - \frac{1}{2} \Delta t \omega_\alpha \left[\frac{e_{\alpha,k} b_k}{c_s^2} \rho + \frac{Q_{\alpha,km}}{c_s^4} \rho u_m b_k \right], \end{aligned} \quad (47)$$

where $Q_{\alpha,km} = e_{\alpha,k}e_{\alpha,m} - c_s^2\delta_{km}$, and $\mathcal{O}(Ma^3)$ and higher terms are neglected in deriving Eq. (47). Thus,

$$\begin{aligned} f_\alpha &= f_\alpha^{(0)} + \varepsilon f_\alpha^{(1)} \\ &= \omega_\alpha \rho \left[1 + \frac{e_{\alpha,k}u_k}{c_s^2} + \frac{Q_{\alpha,km}u_ku_m}{2c_s^4} \right] \\ &\quad - \tau_f \Delta t \omega_\alpha \left[\frac{Q_{\alpha,km}}{c_s^2} \rho \partial_k(u_m) - \frac{e_{\alpha,k}}{c_s^2} \partial_j(\rho u_j u_k) \right. \\ &\quad \left. + e_{\alpha,j} \frac{Q_{\alpha,km}}{2c_s^4} \partial_j(\rho u_k u_m) - \frac{Q_{\alpha,km}}{2c_s^4} \partial_j(\rho u_m u_k u_j) \right] \\ &\quad - \frac{1}{2} \Delta t \omega_\alpha \left[\frac{e_{\alpha,k}b_k}{c_s^2} \rho + \frac{Q_{\alpha,km}}{c_s^4} \rho u_m b_k \right]. \end{aligned} \quad (48)$$

Define $\mathbf{n}$ as the unit normal vector pointing inside the wall, $f_{\bar{\alpha}}$ means the unknown distribution ($e_{\bar{\alpha},j}n_j > 0$), f_α is the distribution of the particles hitting the wall ($e_{\alpha,j}n_j < 0$)

Fixed wall, which means $u_i = 0$ at the wall, the distribution function reduces to:

$$f_{\bar{\alpha}} = \omega_{\bar{\alpha}} \rho - \tau_f \Delta t \omega_{\bar{\alpha}} \frac{Q_{\bar{\alpha},km}}{c_s^2} \rho \partial_k u_m - \frac{1}{2} \Delta t \omega_{\bar{\alpha}} \rho \frac{e_{\bar{\alpha},k}b_k}{c_s^2}. \quad (49)$$

Subtracting f_α from $f_{\bar{\alpha}}$, the no-slip boundary for the mesoscopic distribution function is:

$$f_{\bar{\alpha}} = f_\alpha - \Delta t \omega_{\bar{\alpha}} \rho \frac{e_{\bar{\alpha},k}b_k}{c_s^2} = f_\alpha + \Delta t \omega_{\bar{\alpha}} \rho \frac{e_{\alpha,k}b_k}{c_s^2}. \quad (50)$$

Similarly, the lifting relation for g_α can be derived:

$$\begin{aligned} g_\alpha &= g_\alpha^{(0)} + \varepsilon g_\alpha^{(1)} \\ &= \omega_\alpha \left[T + e_{\alpha,k} \frac{T u_k}{c_s^2} + \frac{Q_{\alpha,km}}{2c_s^4} T u_k u_m \right] \\ &\quad - \tau_g \Delta t \omega_\alpha \left[\frac{Q_{\alpha,km}}{c_s^2} \partial_k(T u_m) - 2Q_{\alpha,km} \frac{T}{\rho} u_m \partial_k p \right. \\ &\quad \left. + e_{\alpha,j} \frac{Q_{\alpha,km}}{2c_s^4} \partial_j(T u_k u_m) - \frac{Q_{\alpha,km}}{2c_s^4} \partial_j(T u_j u_k u_m) \right. \\ &\quad \left. - \frac{e_{\alpha,k}}{c_s^2} \partial_j(T u_j u_k) - \frac{e_{\alpha,k}}{c_s^2} \frac{T}{\rho} \partial_k p + e_{\alpha,j} \partial_j T b_k + 2Q_{\alpha,km} T u_m b_k \right]. \end{aligned} \quad (51)$$

Also assuming that the wall is fixed, we have:

$$\begin{aligned} g_{\bar{\alpha}} &= \omega_{\bar{\alpha}} T - \tau_g \Delta t \omega_{\bar{\alpha}} \left[\frac{Q_{\bar{\alpha},km}}{c_s^2} T \partial_k(u_m) - \frac{e_{\bar{\alpha},k}}{c_s^2} \frac{T}{\rho} \partial_k p \right. \\ &\quad \left. + e_{\bar{\alpha},j} \partial_j T + \frac{e_{\bar{\alpha},k}}{c_s^2} T b_k \right]. \end{aligned} \quad (52)$$

Thus, the following relation for g_α and $g_{\bar{\alpha}}$ can be derived:

$$\begin{aligned} g_\alpha + g_{\bar{\alpha}} &= 2\omega_{\bar{\alpha}} T - 2\tau_g \Delta t \omega_{\bar{\alpha}} \left[\frac{Q_{\bar{\alpha},km}}{c_s^2} T \partial_k(u_m) \right], \\ g_{\bar{\alpha}} - g_\alpha &= -2\tau_g \Delta t \omega_{\bar{\alpha}} \left[-\frac{e_{\bar{\alpha},k}}{c_s^2} \frac{T}{\rho} \partial_k p + e_{\bar{\alpha},j} \partial_j T + \frac{e_{\bar{\alpha},k}}{c_s^2} T b_k \right]. \end{aligned} \quad (53)$$

The addition of g_α and $g_{\bar{\alpha}}$ is related to the temperature. Thus, the anti bounce-back is used for isothermal wall:

$$g_{\bar{\alpha}} = -g_\alpha + 2\omega_{\bar{\alpha}} T. \quad (54)$$

The subtraction of g_α and $g_{\bar{\alpha}}$ is related to heat flux. Thus, the bounce-back is adopted for adiabatic wall:

$$g_{\bar{\alpha}} = g_\alpha - 2\tau_g \Delta t \omega_{\bar{\alpha}} \left[-\frac{e_{\bar{\alpha},k}}{c_s^2} \frac{T}{\rho} \partial_k p + \frac{e_{\bar{\alpha},k}}{c_s^2} T b_k \right]. \quad (55)$$

As the term on the right-hand-side is of high order, the adiabatic boundary condition in LBM usually takes the form:

$$g_{\bar{\alpha}} = g_\alpha. \quad (56)$$

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